

ANUBANDH, explained simply

This note uses easy words. If you can explain it to a 13-year-old, you can explain it to a judge.

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1. What is the problem?

Imagine a school keeps a big notebook of who owns which radio, who is allowed to move it, and who last saw it. The notebook sits on one teacher's desk. If someone erases a line and writes a new one, it looks like the new line was always there. Nobody can prove the old line existed.

Big companies like BEL have the same kind of problem, but with real people and real equipment. Today that notebook is a normal computer database. One person with admin access can change history. Names, keys, and "who held this radio" live in different places. If two people disagree about where the radio is, the system often just keeps the latest guess.

In short, three things go wrong:

- **Anyone with the master password can rewrite the past.**
- **Hiding a button is not real security.** A clever person can still call the system directly.
- **Seeing something is not the same as proving it.** A note that says "radio is in Depot A" can be fake, old, or clash with another note.

The government problem (SIH26125) asks: can we build a software system, on a blockchain, where identity, permission, and assets share one trail that is very hard to secretly edit?

2. What did we build?

ANUBANDH is a website plus a private blockchain. Think of the blockchain as a shared diary. Every important action is a new page. You can add pages. You cannot quietly rip an old page out.

The website is just the window. The diary is the real lock. If a student tries to do a teacher's job, the diary itself says no, even if the student finds a hidden door around the website.

3. How we solve it (simple pieces)

Your name is a lock, not a password

Each person gets a DID. That is a digital name tied to a secret key, like a unique wax seal. To log in you do not type a password. Your key signs a one-time puzzle. The system checks: "Is this still the real key for this name?" If the name was cancelled, login fails.

Jobs have ranks, and the chain enforces them

Rank	Like at school	What they can do here
Admin	Head teacher	Create new items, give ranks, cancel names

Rank	Like at school	What they can do here
Manager	Sports captain	Move items, hand them to someone, suggest a fix after a fight
Auditor	Inspector	Read the diary. Cannot move items.
User	Student on duty	Write a signed note: I saw this item here, in this state

A User who clicks "create item" is refused. That is not only the website being polite. The chain program itself refuses. We even have a test that tries it the sneaky way, skipping the website. It still fails.

Each thing is a unique token

A radio is not just a row in a spreadsheet. It is a unique digital token (an NFT). The token remembers **owner** (who it belongs to) and **custodian** (who is holding it right now). Those two can be different, like owning a cricket bat but lending it to a friend. The long document about the radio stays on a normal computer. Only a fingerprint of that document goes on the chain.

People sign what they saw

When someone checks a radio, they sign a note. There are two kinds of notes:

- **Signed inspection** (strong): "I am allowed to check this, and I sign with my key." Weight 1.0.
- **Identifier scan** (weaker): like scanning a barcode. Easy to copy. Weight 0.4. We never say a QR code cannot be faked.

We also score how much we **trust the current picture**. Fresh signed notes score high. Old notes fade. If two recent notes disagree, trust crashes. The number on the screen always explains itself: which notes, how old, and whether there is a fight.

If two people disagree, freeze the item

User A says the radio is in Depot A and looks fine. User 2, soon after, says Depot B or "broken". The system writes a **divergence**: a permanent record of the clash. Then it **quarantines** the radio. Nobody can transfer it until a manager records an official decision. The fight is never erased. For extra-important items, a manager proposes the fix and a **different** admin must agree. Two people, not one.

Inspectors can rewind time

Every important action leaves a mark in the diary. A helper program copies those marks into a fast search box (Postgres). An auditor can ask: "At 2pm yesterday, who owned this, and was it frozen?" The answer is rebuilt from the marks, not from "whatever the last row says now". If the search box and the diary ever disagree, the diary wins.

It can work a bit without internet

Field staff can save a snapshot on their phone (a PWA). Small checks can still happen offline. Big, dangerous actions (like transferring ownership) are blocked if the saved snapshot is older than 15 minutes. When the network comes back, the signed decisions are sent in and checked again.

4. The path of one click

You in the browser → a gatekeeper program (the PEP) → smart contracts on the chain → a listener copies events → a database for search → the audit screen.

- The website never gets to cheat. The chain has the last word.
- Login is a signature, not a password.
- Local demo chain is Anvil (a practice chain). Real-world target is Hyperledger Besu. Same kind of programs, stricter network.

5. How you show it (no password)

Use the laptop site <http://localhost:3000>. The public site shows the look, but login needs your laptop's backend.

- **Admin:** pick Admin in the list. Click Sign challenge and enter. Show token #1.
- **User cannot mint:** sign out, pick User, try Mint. It should fail. That is the point.
- **Two stories clash:** User signs "Depot A, Good". User 2 signs something different. The radio freezes.
- **Manager:** transfer should fail while frozen. Then propose a fix. The clash stays in history.
- **Auditor:** open Audit Trail and rebuild an earlier moment.

6. What you must not say

True	Not true
The chain records who signed what, and when.	The token proves the real radio is sitting in the depot.
A QR-style scan is a weak clue.	A QR or NFC tag cannot be copied.
The website on Vercel is the front door.	Vercel is the full BEL production system with Besu.
This is a working prototype with real tests.	We already plugged into BEL's live factory software.

7. One line to remember

ANUBANDH does not pretend a computer token is the real world. It makes names, permissions, custody, and disagreements hard to secretly rewrite. The chain is the lock. The website is the window.

Code: github.com/Faizaan-Alam/anubandh-sih26125 | More detail: [ANUBANDH_Presentation_Briefing.pdf](#) in this folder